

What a fibre is in this matrix, and what it is not

prototype/ecm/block_bounce.py · prototype/ecm/block_metrics.py · log/okuda_ECM/02b-02h · 10 August 2026

1. The question, which is about resolution and not about collagen

`seed_ecm` lays the matrix down as strands: 1500 straight segments, each carrying 60 of the 90,000 particles, with a shared orientation and a little across-strand jitter. The movies then show fibres being dragged and splayed, and every statement anyone has been tempted to make from those movies is about *fibres*. The operator's own docstring warns that they are geometry and not mechanics. This note measures whether that warning is right, and it is: the strands are an arrangement of the mass, and the material responds as though they were not there.

Two facts decide it before anything moves, and both are arithmetic on the spec. The background grid is 64^3 , so a cell is $1/64 = 0.0156$ box units. A strand is 0.09 long — 5.8 cells — and its particles sit 0.0015 apart, a *tenth* of a cell, while the jitter that gives the strand its thickness is 0.004, a *quarter* of a cell. So the grid resolves a strand along its length and cannot resolve it across: about ten particles of one strand land in a single cell and are interpolated into one velocity. And no force is ever computed between two particles of the same strand — MLS-MPM computes nothing between two points directly [1, 2], and `seed_ecm` writes each particle's unit fibre vector to an attribute on the operator object that no other operator reads.

A fibre in this model is therefore three things it should not be at once: a line the renderer draws by index arithmetic (`particle i` belongs to `fibre i/per`), a clump of mass the grid smears, and a direction stored where nothing can use it.

2. A load that arrives

Every earlier run loads the matrix the same way — slowly, outward, everywhere at once — so none of them can say whether it carries stress the way a fibrous material should, because there is no moment at which the load appears. A block falling under gravity onto the floor is the cheapest load that *arrives*: the impact is a front travelling up through the strands, and the block either returns the drop or keeps it.

Seven runs, all 1500 strands in a cube of $0.4 \times 0.32 \times 0.4$ box units with a spherical cavity of radius 0.07, 720 frames at $\Delta t = 3.2 \times 10^{-3}$, differing in one number apiece. What is measured is in `block_metrics.py`: the bulk (height, energy, the fraction of the drop each impact returns), the shape (the three dimensions against their own seeded values), and the fibre state — the three things a strand can do that a continuum cannot, which are stretch along itself, bow sideways, and turn.

End-to-end, not contour. The jitter is 0.004 across a strand whose particles are 0.0015 apart, so the contour is a random walk at the particle scale and is six times the end-to-end distance before anything has moved (0.163 at frame 0, and 0.163 again at frame 360). Any per-segment measure reports the seeding noise. End-to-end length, and the RMS distance of a strand's particles from its own end-to-end axis, are both taken over the whole strand, where the jitter averages down.

3. The strands do nothing a continuum would not

Through an impact that compresses the block to 0.914 of its seeded height, a strand's end-to-end length falls to 0.982 of the 0.0896 box units it was seeded with, and returns to 0.999. Its bow goes $0.00824 \rightarrow 0.00827 \rightarrow 0.00826$ box units — an excursion of 0.4% — and the orientation order parameter about the drop axis moves by 0.04. Nothing buckles, nothing stays stretched, nothing turns.

That is not a surprise and it is not a bug: it is the prediction of the constitutive law. `mpm_scatter` computes a fixed-corotated stress from $\mathbf{F}$ alone and reads the strand direction nowhere, so the only anisotropy this matrix can express is *where the mass happens to be*. It is also what the earlier alignment campaign found the expensive way — `align 0.85` about one axis bought a directional pressure difference of $1.50\times$, and a shape only appeared once the fibres were *redistributed* to put three times as many within 55° of the poles, which changes the density the grid sees rather than the direction it responds to.

4. The readout had been the wrong quantity

`ecm_stress` defaulted to `measure: vol`, which is $|J - 1|$, the local volume change — and an impact is a shear. With `store_stress: true` on `mpm_scatter` the solver keeps the Cauchy stress it already computes to build the grid forces, and `measure: vonmises` reads its invariant instead. On the *same trajectory* — the positions are

bit-identical, checked element by element over $721 \times 90,000 \times 3$ floats, so this is a readout and not a mechanism — the two differ by a factor of 250: at frame 200, $|J - 1|$ gives mean 0.0040 and p99 0.0173, the von Mises gives mean 1.11 and p99 4.16 in stress units, peaking at p99 42.4 just after first contact. It costs 45% more wall clock and it is the difference between seeing the front and not. One thing to carry: the run-time `stress_scale` of 2×10^{-5} was chosen for $|J - 1|$ and saturates every band under von Mises; the movie re-bands from the recorded raw values (full scale 9.406, the p99 over the run), but `node_type` in the trajectory is pinned.

5. The block rings, and an extracellular matrix does not

After the impact every shape metric oscillates. The measured period is 24.2 frames, 0.0775 s, 12.9 Hz, against the 17.1 Hz that a free-free bar of length 0.32 at $c = \sqrt{E/\rho} = 10.95$ box units per second would have — so it is the block’s own elastic breathing mode, real physics of the material as specified. It is nonetheless biologically wrong. The stock `drag` of 0.05 is a velocity decay time of $1/\text{drag} = 20$ s, which is 260 ringing periods and 8.7 times the whole 2.31 s run: this matrix is effectively undamped. A hydrated matrix dissipates by solvent forced through the network, and at $\text{Re} \sim 10^{-10}$ it cannot ring at all.

`drag` is the right lever and needs no new operator — it is a Stokes term on the network velocity, which is what Darcy drag is. Raising it costs nothing:

run	<code>drag</code>	ripple, as a fraction of the seeded height	height returned at the first impact	bounding volume at the last frame	median residual displacement (box units)
02c	0.05	0.00813	0.707	1.172	0.03604
02d_drag2	2	0.00475	0.293	0.999	0.00264
02d_drag8	8	0.00054	0.048	0.997	0.00108

The damping that removes the ringing removes the bounce with it: at `drag`= 8 the block returns 4.8% of what it fell. There is no bouncing experiment in the regime where the material is physical, and the three impacts of 02b are a property of an undamped solid rather than of a matrix.

6. What was read as creep, and was not

The undamped runs showed the block slowly inflating — bounding volume rising to 1.34 of its seeded value by frame 560, median particle displacement with the block’s own translation removed reaching 0.036 box units, which is 2.3 grid cells — and I reported that as the one irreversible thing the material did. It is not the material. At `drag`= 2 the volume ends at 0.999 and the residual displacement at 0.0026, smaller by a factor of 13.6; at `drag`= 8, 0.0011. It was undamped particle noise accumulating, and the only reason it looked like creep is that it was monotone.

The general form of that mistake is worth keeping: a quantity that only ever increases is not thereby a material property, and the cheapest test of whether it is one is to change the damping, which no material property should notice.

7. What each lever costs, measured

Four runs, each changing one number from the `drag`= 2 configuration, all 720 frames, solve time timed inside the script and excluding rendering. The last row changes three numbers at once and is the configuration §9 proposes.

run	what changed	particles	substeps per frame	solve (s)	speed-up	first impact (frame)	height returned at that impact	peak compression (of seeded height)
<i>against the <code>drag</code>=2 control</i>								
02d_drag2	—	90,000	16	~ 678*	1.0	181	0.293	0.9468
02e_sub4	$\Delta t_{\text{sub}} 4 \times 10^{-4}$	90,000	8	339.1	2.0	181	0.304	0.9439
02f_sub8	$\Delta t_{\text{sub}} 8 \times 10^{-4}$	90,000	4	180.6	3.8	182	0.314	0.9415
02g_p30k	20 particles per strand	30,000	16	261.9	2.6	181	0.280	0.9443
<i>against the <code>drag</code>=8 control</i>								
02d_drag8	—	90,000	16	~ 678*	1.0	314	0.048	0.9868
02h_lean	all three	30,000	8	115.5	5.9	315	0.043	0.9854

* implied, not timed: the timer was added after those runs. Solve time is exactly linear in substeps at fixed particle count (339.1 s at 8, 180.6 s at 4, i.e. 42.4 and 45.2 s per substep-set), and 02c’s wall clock including compression was 1086 s.

Two levers, and both were being paid for without buying anything. The substep was $\Delta t_{\text{sub}} = 2 \times 10^{-4}$ against a CFL limit of $dx/c = 1.4 \times 10^{-3}$ — a factor of seven of headroom. Halving the substep count moves the peak

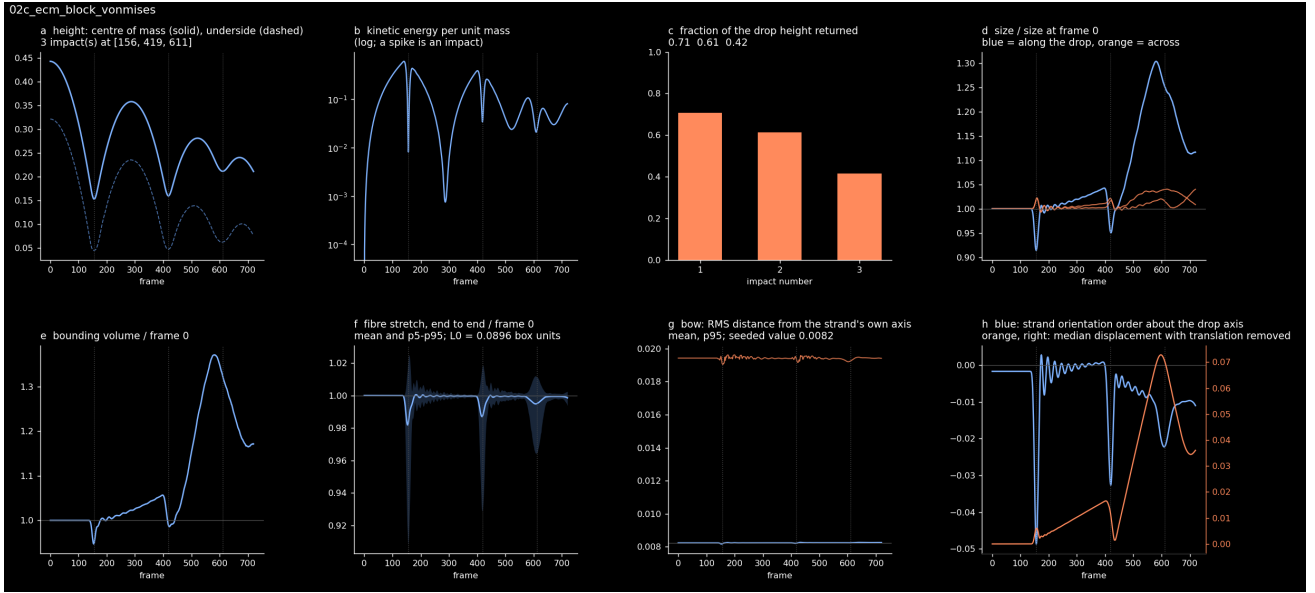


Figure 1: 02c, the undamped block, 721 frames. **a–c** three impacts, and the height returned falling $0.707 \rightarrow 0.612 \rightarrow 0.416$, which reads as dissipation and is in fact the block coming apart — **e** shows its bounding volume growing to 1.34 of the seeded value. **d–e** ring at 12.9 Hz, the block’s own elastic mode. **f–g** are the fibre state and are the point: the end-to-end stretch recovers to 0.999 and the bow moves by 0.4%, so the strands ride the continuum. Everything in **e**, and the orange trace in **h**, is removed by **drag**= 2.

compression by 0.3% and the fibre stretch by 0.01%; quartering it moves them by 0.6% and 0.02% and is still stable, which is why the proposal takes the safe half rather than the quarter. And 60 particles per strand put ten of them in one grid cell, so two thirds of them are describing a strand at a resolution the grid discards: at 20 per strand the peak compression moves by 0.26% and the run is $2.6\times$ cheaper. Combined, the lean configuration reproduces its control to within one frame on the impact, 0.14% on the peak compression and 0.02% on the fibre stretch, at $5.9\times$ less compute.

8. What would falsify a fibre model

Nothing above needs fibres to explain it, so nothing above can falsify a fibre model either. What a real fibre network does, in the order the property matters: it carries load along its axis in tension and buckles under $\sim \pi^2 EI/L^2$ in compression, which for a fibril of aspect ratio 100 is 10^{-4} of its tensile stiffness; that asymmetry is what produces strain stiffening as the network goes from bending-dominated to stretch-dominated near 10–20% strain [3, 4], and long-range transmission, where displacement decays like $r^{-0.5}$ to r^{-1} instead of the r^{-2} of a linear continuum [6]; and its fibres rotate toward the maximum principal stretch, so they end up circumferential around a compressed region.

Three loadings would show it, in increasing cost:

- **Compress between plates to 20% and release.** The tangent modulus at 5% against 20% strain is the stiffening ratio, which a network raises by one to two orders of magnitude and an isotropic solid does not raise at all; the residual strain after unloading is the plasticity; and the effective Poisson ratio separates a solid that bulges (≈ 0.3 from the Lamé pair) from a network that densifies (0 to 0.1).
- **Pull by 10%, push by 10%.** The ratio of secant moduli is 1 by construction for the present law and must exceed 3 for anything that buckles [5].
- **Contract a sphere at the centre by 10% and fit $u(r) \propto r^{-n}$.** An incompressible linear continuum gives $n = 2$, a fibrous matrix n between 0.5 and 1 [6]. That single exponent is the property the whole spheroid–matrix coupling rests on, and it is the test to run first.

One metric belongs in every one of them: the mean $\cos^2 \theta$ between a strand’s axis and the maximum-principal-stress eigenvector of its own particles’ σ . Random gives $1/3$; a fibre-reinforced law drives it toward 1 wherever the strand is in tension. That is “builds stress along its direction” written as a number that can fail, and it needs the one structural change everything else waits on: the fibre direction as a per-particle buffer rather than an attribute on the operator.

9. The spec I propose for the ECM–BM–spheroid study

Keep the two passes. The reason is unchanged and is geometric, not a preference: the MPM grid is fixed to the unit cube, the epithelium lives in a box fifty times larger, and shrinking the tissue to fit changes which term of its energy dominates. What crosses between them is the **surface** Level.

Pass 1, the epithelium, carries no MPM and costs minutes: 200 $\rightarrow$ $\sim$ 1100 cells over 402 frames, **shape_energy_3d** / **reconnect_t1_3d** / **divide_3d**, with **ecm_growth_gate_3d** reading the recorded pressure map [9, 10]. Nothing in this note touches it.

Pass 2, the matrix and the sheet, is where the compute is, and where the four numbers above go:

```
fields: mpm_grid: {n_grid: 64}          # dx = 0.0156; a strand is 5.8 cells long
sets:
  stroma:      {parent: ecm_body, per_parent: 40000, types: {ecm: {youngs: 15}}}
  bm:          node set, NOT mpm_particle # see below
  plaque:      edge set cell -> bm
  fibril:      edge set bm -> ecm
operators:
  - {op: seed_ecm, at: stroma, n_fibres: 6000, fibre_len: 0.16, per_strand: 20}
  - {op: mpm_scatter, at: stroma, drag: 8.0, store_stress: true} # overdamped, keeps sigma
  - {op: ecm_stress, at: stroma, measure: vonmises, scale: <p99 of a calibration frame>}
schedule:
  - {substep_dt: 4.0e-4, steps: [mpm_strain, mpm_scatter, mpm_grid_update, mpm_gather]}
```

Four choices, each with the measurement behind it. **drag**: 8 because at 0.05 the matrix rings for the entire run and accumulates 2.3 grid cells of displacement that is not material (§5, §6), and raising it costs nothing. $\Delta t_{\text{sub}} = 4 \times 10^{-4}$, eight substeps rather than sixteen, because the CFL limit is 1.4×10^{-3} and halving the count moves every deformation measure by less than 0.3% (§7). Twenty particles per strand rather than sixty, because ten of the sixty land in one grid cell (§1) and dropping to twenty moves the same measures by less than 0.3% — for the spheroid’s 6000 strands that is 40,000 particles, not 120,000. And **store_stress/vonmises** because $|J - 1|$ cannot see a shear (§4). Together these are the 5.9 $\times$ of 02h: a pass-2 of about ninety seconds rather than ten minutes, which is what makes a sweep affordable rather than a weekend.

The basement membrane must not be an MPM material, and the reason is a number. A basement membrane is ~ 100 nm thick on a spheroid $\sim 200 \mu\text{m}$ across — one part in two thousand of the radius, which at a spheroid radius of 0.2 box units is 10^{-4} box, or one part in 150 of a grid cell. There is no resolution at which an MPM sheet is a sheet here. Worse, it would set the timestep for everything else: the sheet is 10 to 100 times stiffer than the stroma, and the substep scales as $1/\sqrt{E}$, so **youngs** = 100 $\times$ the stroma’s takes the run from three substeps per frame to twenty-seven — *nine times the cost, to resolve a layer the grid cannot represent*. The sheet therefore stays what §6 of **spheroid_ecm** argues it is: nodes and crosslinks, massless and overdamped, integrated at its own rate, with **plaque** as an edge-set so the adhesion returns its reaction to the cell [11] and **fibril** as an edge-set carrying the stroma’s load through the sheet [12] rather than letting the epithelium push the stroma directly.

Then spend the saving on the one thing that changes an answer. In order, and each is a separate operator with its own falsifier: (i) the fibre direction as a per-particle buffer, which nothing else can proceed without; (ii) a tension-only transversely isotropic term in the stress, $I_4 = |\mathbf{Fa}|^2$ active only when $I_4 > 1$ [7, 8], which is the only way stress can align with a strand, tested by the $\cos^2 \theta$ metric of §8 and by the r^{-n} exponent; (iii) affine reorientation $\mathbf{a} \leftarrow \mathbf{Fa}/|\mathbf{Fa}|$ [6]; (iv) mass balance on the sheet rather than a fixed particle budget [13]. Items (ii) and (iii) are perhaps a 15% surcharge on pass 2 — an I_4 and a rank-one stress per particle per substep — against the 5.9 $\times$ already banked.

One caveat that the saving makes sharper. Under one-way coupling the stroma’s **youngs** is very nearly a free parameter: the tissue never feels the matrix, and the growth gate normalises the pressure map to its own late-run median, so a softer stroma changes the substep it needs and not the shape that comes out. That is convenient — softer is cheaper, since the substep scales as $1/\sqrt{E}$ — and it is also the clearest statement of what one-way coupling costs. The moment an operator writes back to **surface**, **youngs** stops being free, and the numbers in this section have to be re-measured rather than carried over.

10. References

References

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